P.3 The Cartesian Plane and Graphs of Equations

Name _____ Period ____ Date _____

Pre-Calculus P.3 (30-minute plan) · Student sheet

I can…

- use the Distance and Midpoint Formulas.
- find intercepts and test a graph for symmetry.
- write and read the standard equation of a circle.

Vocabulary · fill in the missing word

Intercepts — the points where a graph _____ the x-axis or the y-axis

Symmetry — a graph is symmetric when a _____ maps it onto itself

Standard form of a circle — $(x - h)^2 + (y - k)^2 =$ _____

Part 1 · Notes and examples (fill in each note, then work the example)

Distance: d = _____. **Midpoint:** $((x_1 + x_2)/2, (y_1 + y_2)/2)$

Example 1

Find the distance and the midpoint between (−2, 1) and (3, 4).

Intercepts: for x-intercepts let y = _____; for y-intercepts let x = 0.

Example 2

Find the intercepts of $y = x^2 - 4x$.

Symmetry tests: replace y with −y (x-axis), x with −x (y-axis), or both (the _____).

Circle: $(x - h)^2 + (y - k)^2 =$ _____

Example 3

Write the standard equation of the circle with center (−1, 2) that passes through (3, 4).

Part 2 · Partner practice

Partner practice

Write the equation of the circle with center (2, −3) and radius 4.